

CHAPTER 2

Linear First Order ODEs

1. Use integrating factor method to solve given ODEs.

a) $y' + 3y = t + e^{-2t}$

b) $y' - 2y = t^2 e^{2t}$

c) $y' + y = te^{-t} + 1$

d) $y' + \frac{1}{t}y = 3 \cos(2t), \quad t > 0$

e) $y' - 2y = 3e^t$

f) $ty' - y = t^2 e^{-t}, \quad t > 0$

g) $y' + y = 5 \sin(2t)$

h) $2y' + y = 3t^2$

2. Solve the given IVPs.

a) $y' - y = 2te^{2t}, \quad y(0) = 1$

b) $y' + 2y = te^{-2t}, \quad y(1) = 0$

c) $y' + \frac{2}{t}y = \frac{\cos(t)}{t^2}, \quad y(\pi) = 0 \quad t > 0$

d) $ty' + (t+1)y = t, \quad y(\ln 2) = 1, \quad t > 0$

Classification

3. Determine whether the given equation is separable, linear, neither or both.

a) $\frac{dx}{dt} + xt = e^x$

b) $3t = e^t \frac{dy}{dt} + y \ln t$

c) $3r = \frac{dr}{d\theta} - \theta^3$

d) $x^2 \frac{dy}{dx} + \sin x - y = 0$

e) $(t^2 + 1) \frac{dy}{dt} = yt - y$

f) $x \frac{dx}{dt} + t^2 x = \sin(t)$

Separable Equations

4. Determine whether the given equation is separable.

a) $\frac{dy}{dx} - \sin(x+y) = 0$

- b) $\frac{ds}{dt} = t \ln(s^{2t}) + 8t^2$
 c) $(xy^2 + 3y^2)dy - 2xdx = 0$
 d) $s^2 + \frac{ds}{dt} = \frac{s+1}{st}$
 e) $\frac{dy}{dx} = 4y^2 - 3y + 1$
 f) $\frac{dy}{dx} = \frac{ye^{x+y}}{x^2 + 2}$

5. Solve the given ODE:

- a) $y' = \frac{x^2}{y}$
 b) $y' + y^2 \sin x = 0$
 c) $y' = \cos^2(x) \cos(2y)$
 d) $xy' = \sqrt{1 - y^2}$
 e) $\frac{dy}{dx} = \frac{x - e^{-x}}{y + e^y}$
 f) $\frac{dy}{dx} = \frac{y}{x}$
 g) $\frac{dy}{dx} = \frac{-x}{y}$
 h) $\frac{dx}{dt} = 3xt^2$
 i) $\frac{dy}{dx} = \frac{x}{y^2 \sqrt{1+x}}$
 j) $\frac{dy}{dx} = \frac{\sec^2(y)}{1+x^2}$
 k) $\frac{dx}{dt} - x^3 = x$
 l) $y^{-1}dy + ye^{\cos(x)} \sin(x)dx = 0$
 m) $(x + xy^2)dx + e^{x^2}ydy = 0$
 n) $x \frac{dy}{dx} = \frac{1}{y^3}$
 o) $\frac{dx}{dt} = \frac{t}{xe^{t+2x}}$

6. Find a solution to given IVP and determine, if possible, the interval in which the solution is defined.

- a) $y' = (1 - 2x)y^2, \quad y(0) = -1/6$
 b) $y' = \frac{1 - 2x}{y}, \quad y(1) = -2$
 c) $xdx + ye^{-x}dy = 0, \quad y(0) = 1$
 d) $\frac{dr}{d\theta} = \frac{r^2}{\theta}, \quad r(1) = 2$
 e) $y' = \frac{xy^3}{\sqrt{1+x^2}}, \quad y(0) = 1$
 f) $y' = \frac{2x}{1+2y}, \quad y(2) = 0$
 g) $y' = \frac{3x^2 - e^x}{2y - 5}, \quad y(0) = 1$
 h) $\sin(2x)dx + \cos(3y)dy = 0, \quad y(\pi/2) = \pi/3$
 i) $\sqrt{y}dx + (1+x)dy = 0, \quad y(0) = 1$
 j) $\frac{1}{\theta} \frac{dy}{d\theta} = \frac{y \sin(\theta)}{y^2 + 1}, \quad y(\pi) = 1$

7. Solve the equation

$$\frac{dy}{dx} = \frac{ay + b}{cy + d}$$

where a, b, c and d are constants.

Existence and Uniqueness

8. Determine (without solving the equation) an interval in which the solution of the given initial value problem is certain to exist. *Hint: Use the theorem Existence and Uniqueness, Linear case.*

Pay attention to the form of the equation, if it is not in the form $y' + p(t)y = q(t)$, then first bring it to this form to apply the theorem

- a) $(t - 3)y' + (\ln t)y = 2t, \quad y(1) = 2$
- b) $y' + (\tan t)y = \sin t, \quad y(\pi) = 0$
- c) $(4 - t^2)y' + 2ty = 3t^2, \quad y(-3) = 1$
- d) $y' = \frac{2x}{1 + 2y}, \quad y(2) = 0$
- e) $(\ln t)y' + y = \cot(t), \quad y(2) = 3$

9. State where in the ty -plane the hypothesis of the theorem **Existence and Uniqueness, Nonlinear case** are satisfied.

- a) $y' = \sqrt{1 - t^2 - y^2}$
- b) $y' = \frac{\ln(ty)}{1 - t^2 + y^2}$
- c) $y' + 2ty = (t^2 + y^2)^{3/2}$
- d) $y' = \frac{1 + t^2}{3y - y^2}$

10. Solve the given initial value problem and determine the interval in which the solution exists depends on the initial value y_0 .

- a) $y' = \frac{-4t}{y}, \quad y(0) = y_0$
- b) $y' = 2ty^2, \quad y(0) = y_0$
- c) $y' + y^3 = 0, \quad y(0) = y_0$
- d) $y' = \frac{t^2}{y(1 + t^3)}, \quad y(0) = y_0$

11. By looking at an IVP $dy/dx = f(x, y)$ with $y(x_0) = y_0$, it is not always possible to determine the domain of the solution $y = \phi(x)$ or the interval in which the function $y = \phi(x)$ satisfies the ODE. To see this, consider the following questions.

- a) Solve the ODE $dy/dx = xy^3$.
- b) Give explicitly the solutions to the IVP with $y(0) = 1; y(0) = 1/2; y(0) = 2$.
- c) Determine the domains of the solutions in part (b).
- d) Now solve the ODE with $y(0) = a, a > 0$. Show that the domain approaches to whole real line $(-\infty, \infty)$ as a approaches to 0^+ and the domain shrinks to 0 as a approaches to ∞ .

12. Let $y = y_1(t)$ be a solution of

$$y' + p(t)y = 0,$$

and let $y = y_2(t)$ be a solution of

$$y' + p(t)y = g(t). \tag{1}$$

Show that $y(t) = cy_1(t) + y_2(t)$ is also a solution to the equation (1) for any constant c .

- 13.** (Method of Separation) The method of separation of variables can give us a solution, but it may not give us all solutions. To illustrate this, consider $dy/dx = y^{1/3}$.

a) Use the method of separation to show that

$$y = \left(\frac{2x}{3} + C \right)^{3/2}$$

is a solution.

b) Show that

$$y = \left(\frac{2x}{3} \right)^{3/2}, \quad x > 0$$

solves the IVP

$$\frac{dy}{dx} = y^{1/3}, \quad y(0) = 0.$$

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- c)** Show that the constant function $y \equiv 0$ also satisfies the IVP in part (b). Hence this IVP does not have a unique solution.
- d)** Explain why this does not conflict with the theorem **Existence and Uniqueness, Nonlinear case**.
- 14.** (Method of Separation) To apply method of separation, we might need to divide the given equation by a function of the dependent variable and by dividing we might lose some solutions. To explore this further:
- a)** Use method of separation to solve

$$\frac{dy}{dx} = (x - 3)(y + 1)^{2/3}.$$

- b)** Show that the constant function $y \equiv -1$ also satisfies the above equation.
- c)** Show that there is no constant c that will make the solution in part (a) yield the solution $y \equiv -1$. Thus, we lost the solution $y \equiv -1$ when we divided by $(y + 1)^{2/3}$ to separate variables.

Substitution

- 15.** Sometimes it is possible to solve a nonlinear equation by making a change of variable of the dependent variable that converts it into a linear equation. One important example of such equations is **Bernoulli equation**:

$$y' + p(t)y = q(t)y^n. \quad (2)$$

Solve the following questions about this kind of equations.

- a)** Solve Bernoulli's equation when $n = 0$ and $n = 1$.
- b)** Show that if $n \neq 0$, then the substitution $v = y^{1-n}$ reduces the Bernoulli's equation to a linear equation in the function v .

- c) Use the method described in part (b) to solve

$$y' = y - y^2.$$

Verify the function you found solves the given equation.

- d) Use the method described in part (b) to solve

$$y' + 2y = xy^{-2}.$$

Verify the function you found solves the given equation.

Exact Equations & Method of Integrating Factor

16. Determine whether each of the given equation is exact. If it is exact, find the solution.

- a) $(2x + 3) + (2y - 2)y' = 0$
- b) $(2x + 4y) + (2x - 2y)y' = 0$
- c) $(3x^2 - 2xy + 2) + (6y^2 - x^2 + 3)y' = 0$
- d) $\frac{dy}{dx} = -\frac{ax + by}{bx + cy}$, a, b, c are constants.
- e) $\frac{dy}{dx} = -\frac{ax - by}{bx - cy}$, a, b, c are constants.
- f) $(ye^{xy} \cos(2x) - 2e^{xy} \sin(2x) + 2x) + (xe^{xy} \cos(2x) - 3)y' = 0$
- g) $\left(\frac{y}{x} + 6x\right) + (\ln(x) - 2)y' = 0$, $x > 0$
- h) $\frac{y}{(x^2 + y^2)^{3/2}} + \frac{x}{(x^2 + y^2)^{3/2}} \frac{dy}{dx} = 0$

17. Identify each equation as separable, linear, exact, or having an integrating factor that is a function of either x alone or y alone.

- a) $(2x + yx^{-1})dx + (xy - 1)dy = 0$
- b) $(2y^3 + 2y^2)dx + (3y^2x + 2xy)dy = 0$
- c) $(2x + y)dx + (x - 2y)dy = 0$
- d) $(y^2 + 2xy)dx - x^2dy = 0$
- e) $(x^2 \sin x + 4y)dx + xdy = 0$
- f) $(2y^2x - y)dx + xdy = 0$

18. Solve the given IVP:

- a) $(2x - y) + (2y - x)y' = 0$, $y(1) = 3$
- b) $(9x^2 + y - 1) - (4y - x)y' = 0$, $y(1) = 0$

19. For each of the following equations find the value of b that makes the equation exact, and then solve it using that value of b .

- a) $(xy^2 + bx^2y) + (x^3 + x^2y)y' = 0$

b) $(ye^{2xy} + x) + bxe^{2xy}y' = 0, \quad y(1) = 0$

20. Find an integrating factor of the form $x^n y^m$ and solve the equation.

a) $(2y^2 - 6xy)dx + (3xy - 4x^2)dy = 0$

b) $(12 + 5xy)dx + (6xy^{-1} + 3x^2)dy = 0.$

21. Show that any separable equation

$$M(x) + N(y)y' = 0$$

is also exact.

22. Find an integrating factor and solve the given equation.

a) $(3x^2y + 2xy + y^3) + (x^2 + y^2)y' = 0$

b) $y' = e^{2x} + y - 1$

c) $1 + (x/y - \sin(y))y' = 0$

d) $y + (2xy - e^{-2y})y' = 0$

23. Show that the given equation is not exact but becomes exact when multiplied by the given integrating factor. Then solve the equation.

a) $(x^2 + y^3) + x(1 + y^2)y' = 0, \quad \mu(x, y) = 1/(xy^3)$

b) $(x + 2)\sin(y) + (x \cos(y))y' = 0, \quad \mu(x, y) = xe^x$

c) $(3xy + y^2) + (x^2 + xy)y' = 0$

24. a) Prove that $Mdx + Ndy = 0$ has an integrating factor that depends only on the sum $x + y$ if and only if the expression

$$\frac{N_x - M_y}{M - N}$$

depends only on $x + y$.

b) Use part (a) to solve the equation

$$(3 + y + xy)dx + (3 + x + xy)dy = 0.$$

25. Verify that when the linear ODE

$$(p(x)y - q(x))dx + dy = 0$$

is multiplied by $\mu(x) = e^{\int p(x)dx}$, the result is exact.

CHAPTER 3

Homogeneous, Constant Coefficient

1. Find the general solution to given ODE.

- a) $y'' + 2y' - 3y = 0$
- b) $y'' + 3y' + 2y = 0$
- c) $y'' + 5y' = 0$
- d) $4y'' - 9y = 0$
- e) $y'' - 2y' + 2y = 0$
- f) $y'' - 2y' + 6y = 0$
- g) $y'' + 2y' + 2y = 0$
- h) $y'' + 6y' + 13y = 0$
- i) $9y'' + 9y' - 4y = 0$
- j) $y'' - 2y' + y = 0$
- k) $16y'' + 24y' + 9y = 0$
- l) $y'' - 6y' + 9y = 0$
- m) $9y'' + 6y' + y = 0$

2. Find the solution to given IVP. Describe the behaviour of the solution as $t \rightarrow \infty$.

- a) $y'' + y' - 2y = 0$; $y(0) = 1$, $y'(0) = 1$
- b) $2y'' + y' - 4y = 0$; $y(0) = 0$, $y'(0) = 1$
- c) $y'' + 3y' = 0$; $y(0) = -2$, $y'(0) = 3$
- d) $4y'' - y = 0$; $y(-2) = 1$, $y'(-2) = -1$
- e) $y'' + 4y = 0$; $y(0) = 0$, $y'(0) = 1$
- f) $y'' - 2y' + 5y = 0$; $y(\pi/2) = 0$, $y'(\pi/2) = 2$
- g) $y'' + y = 0$; $y(\pi/3) = 2$, $y'(\pi/3) = -4$
- h) $y'' + 2y' + 2y = 0$; $y(\pi/4) = 2$, $y'(\pi/4) = -2$
- i) $9y'' - 12y' + 4y = 0$; $y(0) = 2$, $y'(0) = -1$
- j) $y'' - 6y' + 9y = 0$; $y(0) = 0$, $y'(0) = 2$
- k) $y'' + 4y' + 4y = 0$; $y(-1) = 2$, $y'(-1) = 1$

3. Find the ODE whose general solution is $y = c_1 e^{2t} + c_2 e^{-3t}$.

4. Consider the equation $ay'' + by' + cy = 0$, where a, b, c are constants with $a > 0$. Find conditions on a, b and c such that the roots of the characteristic equation are:

- (i) real, different, and negative.
- (ii) real with opposite sign.
- (iii) real, different and positive.

In each case, determine the behaviour of the solution as $t \rightarrow \infty$.

Wronskian, Superposition, Linear Independence

5. Find the Wronskian of the given pair of functions.

- a) $e^{2t}, e^{-3t/2}$
- b) $\cos t, \sin t$
- c) e^{-2t}, te^{-2t}
- d) $e^t \sin t, e^t \cos t$
- e) $\cos^2 \theta, 1 + \cos(2\theta)$
- f) $e^{\lambda t} \cos(\mu t), e^{\lambda t} \sin(\mu t)$ for μ, λ real numbers.

6. Verify that $y_1(t) = t^2$ and $y_2(t) = t^{-1}$ are two solutions of the differential equation $t^2 y'' - 2y = 0$ for $t > 0$. Then show that $y = c_1 t^2 + c_2 t^{-1}$ is also a solution of this equation for any c_1 and c_2 .
7. If $W[f, g](t) = t \cos t - \sin t$, and if $u = f + 3g$, $v = f - g$, find $W[u, v](t)$.
8. Verify that $y_1(t) = 1$ and $y_2(t) = t^{1/2}$ are solutions of the differential equation $yy'' + (y')^2 = 0$ for $t > 0$. Then show that $y = c_1 + c_2 t^{1/2}$ is not, in general, a solution of this equation. Explain why this result does not contradict with *Principle of Superposition Theorem*.
9. Show that if $y = \phi(t)$ is a solution of the differential equation $y'' + p(t)y' + q(t)y = g(t)$, where $g(t)$ is not always zero, then $y = c\phi(t)$, where c is any constant other than 1, is not a solution. Explain why this result does not contradict with *Principle of Superposition Theorem*.
10. Find the Wronskian of two solutions of the given differential equations without solving the equation. Hint: Use *Abel's theorem*.
- $t^2 y'' - t(t+2)y' + (t+1)y = 0$
 - $(\cos t)y'' + (\sin t)y' - ty = 0$
 - $(1-x^2)y'' - 2xy' + \alpha(\alpha+1)y = 0$
11. Verify that the functions y_1 and y_2 are solutions of the given differential equation. Do they constitute a fundamental set of solutions?
- $y'' + 4y = 0$; $y_1(t) = \cos(2t)$, $y_2(t) = \sin(2t)$
 - $y'' - 2y' + y = 0$; $y_1(t) = e^t$, $y_2(t) = te^t$
 - $x^2 y'' - x(x+2)y' + (x+2)y = 0$; $x > 0$; $y_1(x) = x$, $y_2(x) = xe^x$
12. Consider the equation $y'' - y' - 2y = 0$.
- Show that $y_1(t) = e^{-t}$ and $y_2(t) = e^{2t}$ form a fundamental set of solutions.
 - Let $y_3(t) = -2y_1(t)$, $y_4(t) = y_1(t) + 2y_2(t)$ and $y_5(t) = 2y_1(t) - 2y_3(t)$. Are $y_3(t)$, $y_4(t)$ and $y_5(t)$ also solutions of the given differential equations? Verify your answer.
 - Determine whether each of the following pairs form a fundamental set of solutions: $\{y_1(t), y_3(t)\}$; $\{y_2(t), y_3(t)\}$; $\{y_1(t), y_4(t)\}$; $\{y_4(t), y_5(t)\}$. Explain.
13. Show that if p is differentiable and $p(t) > 0$, then the Wronskian $W(t)$ of two solutions of $(p(t)y')' + q(t)y = 0$ is $c/p(t)$ where c is a constant.
14. Show that $y_1(t) = \cos t$ and $y_2(t) = \sin t$ are a fundamental set of solutions of $y'' + y = 0$; that is, show that they are solutions and their Wronskian is not zero.
15. **(Alternate way to find the second solution)** Consider the equation $y'' + 2ay' + a^2y = 0$.
- Show that the roots of the characteristic polynomial are $r_1 = r_2 = -a$ so that one solution of the equation is e^{-at} .
 - Use Abel's formula to show that the Wronskian of any two solutions of the given equation is

$$W[y_1, y_2](t) = y_1(t)y_2'(t) - y_1'(t)y_2(t) = c_1 e^{-2at},$$

where c_1 is a constant.

- c) Let $y_1(t) = e^{at}$ and use the result on part (b) to obtain a differential equation satisfied by the second solution $y_2(t)$. By solving this equation, show that $y_2(t) = te^{at}$.

16. Use the method described in the question (15.) or (**Alternate way to find the second solution**) to find a second independent solution of the given equation.

- a) $t^2y'' + 3ty' + y = 0, t > 0; \quad y_1(t) = t^{-1}$
b) $ty'' - y' + 4t^3y = 0, t > 0; \quad y_1(t) = \sin(t^2)$

Reduction of Order

17. Use the method of reduction of order to find a second solution of the given differential equation.

- a) $t^2y'' - 4ty' + 6y = 0, t > 0; \quad y_1(t) = t^2$
b) $t^2y'' + 2ty' - 2y = 0, t > 0; \quad y_1(t) = t$
c) $t^2y'' + 3ty' + y = 0, t > 0; \quad y_1(t) = t^{-1}$
d) $ty'' - y' + 4t^3y = 0, t > 0; \quad y_1(t) = \sin(t^2)$

Substitution

18. Apply the substitution $x = \ln t$ to solve following equations:

- a) $t^2y'' + ty' + y = 0; t > 0$
b) $t^2y'' + 4ty' + 2y = 0; t > 0$
c) $t^2y'' + 7ty' + 10y = 0; t > 0$
d) $4t^2y'' - 8ty' + 9y = 0; t > 0$

Existence and Uniqueness, Linear, Homogeneous

19. Determine the largest interval in which the given initial value problem is certain to have a unique twice differentiable solution. Explain why. Do not attempt to find the solution.

- a) $y'' + (\cos t)y' + 3(\ln |t|)y = 0, \quad y(2) = 3, \quad y'(2) = 1$
b) $(x - 2)y'' + y' + (x - 2)(\tan x)y = 0, \quad y(3) = 1, \quad y'(3) = 2$

Nonhomogeneous Equations

Method of Undetermined Coefficients

20. Decide whether or not the method of undetermined coefficients can be applied to find a particular solution of the given equation.

- a) $y'' + 2y' - y = t^{-1}e^t$
b) $5y'' - 3y' + 2y = -t^3 \cos(4t)$
c) $2y'' - 6y' + y = \sin x / e^{4x}$
d) $x'' + 5x' - 3x = 3^t$
e) $y'' + 3y' - y = \sec(\theta)$
f) $y'' + 2y' + y = x^{100}e^{4x} \cos(25x)$

21. Find the general solution of the given ODE by using method of undetermined coefficients.

- | | |
|--|---|
| a) $y'' - 2y' - 3y = 3e^{2t}$ | g) $y'' + y = 3\sin(2t) + t\cos(2t)$ |
| b) $y'' - y' - 2y = -2t + 4t^2$ | h) $u'' + \omega_0^2 u = \cos(\omega t), \quad \omega^2 \neq \omega_0^2$ |
| c) $y'' + y' - 6y = 12e^{3t} + 12e^{-2t}$ | i) $u'' + \omega_0^2 u = \cos(\omega_0 t)$ |
| d) $y'' - 2y' - 3y = -3te^{-t}$ | j) $y'' + y' + 4y = 2\sinh t$ |
| e) $y'' + 2y' = 3 + 4\sin(2t)$ | k) $y'' - y' + y = (e^t + t)^2$ |
| f) $y'' + 2y' + y = 2e^{-t}$ | l) $3y'' + 2y' + 8y = t^2 + 4t - t^2 e^t \sin t$ |

22. Find the form for a particular solution to

- a)** $y'' + 2y' - 3y = g(t)$
b) $y'' - 2y' + y = g(t)$

where $g(t)$ is equals to

- i)** $7\cos(3t)$ **ii)** $2te^t \sin t$ **iii)** $t^2 \cos(\pi t)$ **iv)** $5e^{-3t}$ **v)** $3te^t$ **vi)** $t^2 e^t$

23. Find the solution to the given IVP by using method of undetermined coefficients.

- a)** $y'' + y' - 2y = 2t; \quad y(0) = 0, \quad y'(0) = 1$
b) $y'' + 4y = t^2 + 3e^t; \quad y(0) = 0, \quad y'(0) = 2$
c) $y'' - 2y' + y = te^t + 4; \quad y(0) = 1, \quad y'(0) = 1$
d) $y'' + 4y = 3\sin(2t); \quad y(0) = 2, \quad y'(0) = -1$
e) $y'' + 2y' + 5y = 4e^{-t} \cos(2t); \quad y(0) = 1, \quad y'(0) = 0$

24. Determine a suitable form of the particular solution, $Y(t)$, of the given ODE by using method of undetermined coefficients. You are suggested to use a computer algebra system to find the coefficients. (e.g: WolframAlpha)

- a)** $y'' + 3y' = 2t^4 + t^2 e^{-3t} + \sin(3t)$
b) $y'' - 5y' + 6y = e^t \cos(2t) + e^{2t}(3t + 4) \sin t$
c) $y'' + 2y' + 2y = 3e^{-t} + 2e^{-t} \cos t + 4e^{-t} t^2 \sin t$
d) $y'' + 4y = t^2 \sin(2t) + (6t + 7) \cos(2t)$
e) $y'' + 3y' + 2y = e^t(t^2 + 1) \sin(2t) + 3e^{-t} \cos t + 4e^t$
f) $y'' + 2y' + 5y = 3te^{-t} \cos(2t) - 2te^{-2t} \cos t$

25. Consider the equation

$$y'' - 3y' - 4y = 2e^{-t}.$$

Note that $y_1(t) = e^{-t}$ and $y_2(t) = e^{4t}$ are solutions of the corresponding homogeneous equation. Adapting the method of reduction of order, we will seek for a solution of the nonhomogeneous equation of the form $v(t)e^{-t}$. To this end, let $Y(t) = v(t)e^{-t}$.

- a)** Substitute $Y(t)$, $Y'(t)$ and $Y''(t)$ into given equation to find an ODE v must satisfy.
b) Use the substitution $w(t) = v'(t)$, to solve the ODE you found in part (a) solving for w , and integrating w to find v .

Method of Variation of Parameters

- 26.** Find a particular solution first by undetermined coefficients and then by variation of parameters. Which method was quicker?

- a) $y'' - 5y' + 6y = 2e^t$
- b) $y'' - y' - 2y = 2e^{-t}$
- c) $4y'' - 4y' + y = 16e^{t/2}$
- d) $y'' - y = 2t + 4$
- e) $2x''(t) - 2x'(t) - 4x(t) = 2e^{2t}$

- 27.** Find the general solution to given differential equation.

- a) $y'' + y = \tan t, \quad 0 < t < \pi/2$
- b) $y'' + 9y = 9 \sec^2(3t), \quad 0 < t < \pi/6$
- c) $y'' + 4y' + 4y = t^{-2t}e^{-2t}, \quad t > 0$
- d) $4y'' + y = 2 \sec(t/2), \quad -\pi < t < \pi$
- e) $y'' - 2y' + y = e^t/(1 + t^2)$
- f) $y''(\theta) + y(\theta) = \sec^3(\theta)$
- g) $v'' + 4v = \sec^4(2t)$
- h) $y'' - 5y' + 6y = g(t)$, where g is continuous function.

- 28.** Verify that the given functions y_1 and y_2 satisfy the corresponding homogeneous equation; then find a particular solution of the given nonhomogeneous equation.

- a) $t^2y'' - 2y = 3t^2 - 1, \quad t > 0; \quad y_1(t) = t^2, \quad y_2(t) = t^{-1}$
- b) $t^2y'' - t(t+2)y' + (t+2)y = 2t^3, \quad t > 0; \quad y_1(t) = t, \quad y_2(t) = te^t$
- c) $ty'' - (1+t)y' + y = t^2e^{2t}, \quad t > 0; \quad y_1(t) = 1+t, \quad y_2(t) = e^t$
- d) $x^2y'' - 3xy' + 4y = x^2 \ln x, \quad x > 0; \quad y_1(x) = x^2, \quad y_2(x) = x^2 \ln x$
- e) $x^2y'' + xy' + (x^2 - \frac{1}{4})y = 3x^{3/2} \sin x, \quad x > 0; \quad y_1(x) = x^{-1/2}, \quad y_2(x) = x^{-1/2} \cos x$

- 29.** a) By choosing the the lower limit of integration in the Variation of Parameters theorem as the initial point t_0 , show that $Y(t)$ becomes

$$Y(t) = \int_{t_0}^t \frac{y_1(s)y_2(t) - y_1(t)y_2(s)}{y_1(s)y_2'(s) - y_1'(s)y_2(s)} g(s) ds.$$

Show that $Y(t)$ is a solution of the initial value problem

$$L[y] = g(t), \quad y(t_0) = 0, \quad y'(t_0) = 0.$$

- b) Show that the solution of the initial value problem

$$L[y] = y'' + p(t)y' + q(t)y = g(t), \quad y(t_0) = y_0, \quad y'(t_0) = y'_0$$

can be written as $y = u + v$, where u and v are solutions of the two initial value problems

$$\begin{aligned} L[u] &= 0, \quad y(t_0) = y_0, \quad y'(t_0) = y'_0 \\ L[v] &= g(t), \quad y(t_0) = 0, \quad y'(t_0) = 0 \end{aligned}$$

respectively. In other words, the nonhomogeneities in the differential equation and in the initial conditions can be dealt with separately. Observe that u is easy to find if a set of fundamental set of solutions of $L[u] = 0$ is known. And, the formula for the function v is given in part (a).

- c)** Use part (a) and (b) to solve following IVP:

$$y'' + y = g(t), \quad y(0) = y_0, \quad y'(0) = y'_0,$$

where g is an arbitrary continuous function.

- 30.** The method of reduction of order can also be used for the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t),$$

provided one solution y_1 of the corresponding homogeneous equation is known.

- a)** Let $y = v(t)y_1(t)$ and show that y satisfies above equation if v is a solution of

$$y_1(t)v'' + (2y_1'(t) + p(t)y_1(t))v' = g(t).$$

Note that this equation is first order in v' , so we can solve for v' and integrate the result to find v . Multiplying y_1 with v gives us the general solution. So, this method simultaneously finds both the second homogeneous equation and a particular solution.

- b)** Using the method introduced in part (a), solve

$$t^2y'' - 2ty' + 2y = 4t^2, \quad t > 0; \quad y_1(t) = t.$$

- c)** Using the method introduced in part (a), solve

$$t^2y'' + 7ty' + 5y = t, \quad t > 0; \quad y_1(t) = t^{-1}.$$

- d)** Using the method introduced in part (a), solve

$$ty'' - (1+t)y' + y = t^2e^{2t}, \quad t > 0; \quad y_1(t) = 1 + t.$$

CHAPTER 4

Independence

- Determine whether the given functions are linearly dependent or independent. If they are linearly dependent, find a linear relation among them.
 - $\phi_1(t) = 2t - 3, \phi_2(t) = t^2 + 1, \phi_3(t) = 2t^2 - t$
 - $\phi_1(t) = 2t - 3, \phi_2(t) = 2t^2 + 1, \phi_3(t) = 3t^2 + t$
 - $\phi_1(t) = 2t - 3, \phi_2(t) = t^2 + 1, \phi_3(t) = 2t^2 - t, \phi_4(t) = t^2 + t + 1$
- Show that $W[5, \sin^2 t, \cos(2t)] = 0$ for all t by directly evaluating the Wronskian.
 - Establish the same result without direct evaluation of the Wronskian.